

problems/ — Onboarding Overview

A reader’s guide to the eight problem subfolders under **problems/**. Each section is self-contained: the conjecture, the current status, the folder layout, the files to open first, and the artifacts worth knowing about. Compiled 2026-05-17 from per-folder reconnaissance by a team of reviewer agents.

At-a-glance index

Subfolder	Conjecture / topic	State	Headline artifact
3_decomposition_coloring/	Hoffmann/ Ostenhof 3-decomposition of cubic graphs	Open; structural reductions advancing through n 14	data/n14_full.summary.json
arc_disjoint_strongly_spanning_Subgraphs/	Engel’s biplanar SAD conjecture (WC3: K = 3)	Open; structural reductions advancing through n 14	paper/draft_v1.md, team/04_ec_log_proof.md
crossing_numbers_Alderson’s/	Alderson’s conjecture (cr(G) cr(K_t) when (G) t)	Open at t 25; three finished side-papers	deliverables/D8_paper/, D15_.../, D16_.../
directed_path_minimum_outdegree/	Chung-Kuo Thomassé directed-path conjecture	Active; = 3 and = 4 for n 11 closed	docs/k3_hand_proof.md, docs/k4_n11_proof.md
earth_moon_problem/	Engel’s biplanar chromatic number _EM [9, 12]	Open; Fork C probe ruled out one closure path	docs/phase6_discharge_attempt.md
pebbling_cartesian_products/	Grundy’s pebbling conjecture on the Lemke square	Terminal; (L L) 246 (rooted 106)	docs/terminal_report.md
positive_square_end_degree_equality/	Alameddine/ Mohar–Pragada– Zhang Conjecture 9.2 for 2-trees	Open at the “slot-shift” wall; 40-page paper drafted	paper/paper.pdf
unit_vector_flows/	Jain’s S ² -flow conjecture for bridgeless graphs	Finite theorem proved for snarks on n 28	THEOREM.md, paper/main.tex

How to read this document

Each section below follows the same five-part shape:

1. **Conjecture** — the precise mathematical statement.
2. **Status** — what is proved, what was refuted, what is still open here.
3. **Layout** — one line per top-level subfolder.
4. **Start here** — 2–4 ordered file paths to open first.
5. **Notable artifacts** — papers, certificates, key scripts, recorded negative results.

If you only have ten minutes, the top of each section’s “Start here” list is the right entry point.

1. 3_decomposition_conjecture/

Conjecture

Hoffmann-Ostenhof (2011), OPG (2017). Every connected cubic graph G has a decomposition $E(G) = T \cup C \cup M$ where T is a spanning tree, C is a 2-regular subgraph (disjoint union of cycles), and M is a matching. Both C and M may be empty. Equivalently, vertices split into V_C (T-degree 1, on a cycle of C), V_M (T-degree 2, matched by M), V_T (T-degree 3, pure tree), with $|V_C| = |V_T| + 2$.

Status

Open. Partial reductions are in hand:

- **Lemma 1 (bridge reduction)** proved; sub-lemma 1’ verified computationally on subcubic graphs up to $n = 11$.
- **Antichain Coverage Conjecture refuted at $n = 12$** (10 out of 1670 oriented 2-pole sides cannot be replaced by gadgets in the $n = 10$ lattice); failures are absorption-universal and handled via compatibility replacement.
- **$n = 14$ sweep complete:** 15,176 of 15,178 oriented 2-pole sides are trace-contained in the $n = 12$ lattice; the remaining 2 are absorbed by the bridge lemma.
- **Live gap:** the Universal Replacement Conjecture — every 2-edge-cut side of order ≤ 12 is either trace-contained by a smaller gadget or compatibility-universal.

Layout

- **docs/** — strategic plan, minimal-counterexample theory, proof logs, literature.
- **scripts/** — verifiers (`decomposition.py`), SAT interface, lattice builder, sweeps.
- **data/** — computed gadget lattices and failure classifications (JSON).

- `tests/` — 33 regression tests (K , K , , Petersen, prism, trace sets).
- `external/` — placeholder.

Start here

1. `README.md` — current status and key results.
2. `docs/plan.md` — vertex-type reformulation, reductions, 7-phase attack plan.
3. `docs/minimal_counterexample.md` §1–3 — boundary-trace formalism + bridge lemma.
4. `scripts/decomposition.py` — trust-root verifier and brute-force finder.

Notable artifacts

- `data/gadget_lattice_2pole_n12_both.json` — 274 oriented gadgets, 59 trace classes.
- `data/failure_structural_classification_n12.json` — structural breakdown of the 10 $n = 12$ failures.
- `data/n14_full.summary.json` — traces for all 15,178 $n = 14$ oriented sides.
- `scripts/full_replacement_sweep.py` — reproducible, resumable sweep driver.

2. `arc_disjoint_strong_spanning_subdigraphs/`

Conjecture

Bang-Jensen–Yeo (2004). There exists an absolute constant K such that every K -arc-strong digraph admits a *strong arc decomposition* (SAD): a 2-coloring of arcs into two color classes each spanning and strongly connected.

Working Conjecture (WC3) in this project: $K = 3$. All known obstructions sit at arc-connectivity 2 — S , squares of even directed cycles, the four semi-complete exceptions of Bang-Jensen–Gutin–Yeo (2020), and the 2-arc-strong split exceptions of Ai–He–Li–Qin–Wang (2024).

Status

Open. Phase 3 v2 closed with a meaningful negative result and two side-theorems:

- **Negative search:** 4,613 verified 3-arc-strong digraphs generated across four vehicles (template gluings, Eulerian constructions, laminar cut systems); zero UNSAT on both ILP and SAT backends. Either WC3 holds, or a counterexample lies outside the structured families tested at $n \leq 30$.
- **Theorem (EC-log).** Every Eulerian digraph D with $|D| \leq 5 \log n$ admits a SAD. Proof via reduction to Karger cut counting + first-moment method.

- **Theorem (CL1, controlled-lifting lemma).** Bilateral SAD decomposition for near-split digraphs.

Layout

- `code/` — cross-checked ILP (PuLP+CBC) and SAT (pysat+CaDiCaL) verifiers; generator modules (`glue.py`, `eulerian.py`, `laminar.py`); benchmark suite.
- `paper/` — `draft_v1.md`, `findings.md`, `outline.md`, `review_v1.md`.
- `team/` — 33 agent outputs (charter, structural program, verifier design, EC-log proof, phase reports v1/v2, lifting-lemma proofs, hard-case audits, recoloring, termination, F3 verification, parallel-closure status).

Start here

1. `attack_plan.md` — roadmap, WC3 definition, EC-log statement, phased milestones.
2. `review.md` — auditor’s sign-off on EC-log and literature corrections.
3. `paper/findings.md` — honest knowledge state: what is proved vs. open.
4. `team/07_phase3_report_v2.md` — phase 3 results in full (4,613 instances, vehicles, stop conditions).

Notable artifacts

- `team/04_ec_log_proof.md` — EC-log proof.
- `team/11_cl1_proof_v1.md` — CL1 lifting-lemma proof.
- Validation suite (`code/`): $\mathcal{S}$, cycle squares, semicomplete and split exceptions as UNSAT benchmarks; small SAT cases.
- `code/run_phase3_v2.py` — explicit families enumerated in the negative search.

3. crossing_numbers_and_coloring/

Conjecture

Albertson (2007). For every graph G with chromatic number $\chi(G) = t$, $\text{cr}(G) \geq \text{cr}(K_t)$. A metric strengthening in the lineage of Hadwiger and (refuted) Hajós.

Status

Open at $t = 25$. Cranston (2025, arXiv:2512.08020) closes $t = 24$ via repeated applications of the Crossing Lemma.

- **Track A (residual hunt).** Cranston’s Theorem 2 isolates pairs $(t, |G|) \in \{(25, 48), (26, 50), (26, 51)\}$ as the only possible counterexamples. The Ore-order congruence $|V(G)| \equiv 1 \pmod{k-1}$ eliminates

(25,48) and (26,50) from the Ore-critical family; only (26,51) survives. Demoted from the headline target.

- **Track B (structural).** R5a — sharpness of Claim 3.7 in Fox–Pach–Suk — closed on 2026-05-17. D8 proves the FPS degree threshold $\tau = 9/8$ is sharp within the Vizing–Gupta + semi-random construction, via the witness identity $f_{\{2b\}}(4/7, \tau) - 9/16 = 12(\tau - 9/8)^2 / [7(4 - 1)]$. The constant $9/16$ is binding in this framework. New front-runners: R2c (min-degree-refined Crossing Lemma) and R3.6 (fractional / list / DP Albertson).

Layout

- docs/ — planning, reviews, literature audit.
- work/ — nine roles (teams) with per-role memos.
- deliverables/ — published papers and supporting computation, indexed D2–D18.

Deliverable highlights:

- D2_literature_verification.md — audit of Cranston, Kostochka–Yancey, Büngener–Kaufmann thresholds.
- D3_R5a_reconstruction.md — faithful reconstruction of FPS Claim 3.7.
- D4_ore_26_51/ — Ore-graph enumeration at (26,51).
- D5_sympy_freedelta/ — SymPy verification of Case 2b monotonicity / witness identity.
- D8_paper/ — sharpness_9_8.pdf (7 pages, theorem-grade); plus the retracted tighter_fps_RETRACTED.pdf with preserved error analysis.
- D12_ore_c3/ — Ore composition tables; lower bound for the C3 construction.
- D13_r2c_attack/, D14_r36_attack/ — in-progress attack memos.
- D15_list_albertson_paper/ — list_albertson_le_18.pdf (9 pages): list-chromatic Albertson for $\tau = 18$.
- D16_expander_crossing_paper/ — expander_crossing.pdf (9 pages): explicit bisection-width Crossing Lemma for spectral expanders.
- D17_submission_packets/ — bundling and journal recommendations.
- D18_combined_observations/ — withdrawn (false observation at $\tau = 5$); preserved.

Start here

1. docs/plan.md v4 — 12-month roadmap; six routes, three tracks, four open questions. Begin at “Honest tractability verdict”.
2. docs/review_v3.md — senior post-team audit; the “Main correction” kills the false (25,48) Ore corner.
3. deliverables/D8_paper/README.md — R5a context, witness identity, retraction notes.

4. `work/01_principal_lead/INTEGRATION.md` — role contract, R-route ownership, R5a closeout + next steps.

Notable artifacts

- Closed: `D8_sharpness_9_8.pdf` (theorem-grade).
- Published-ready: D15 and D16 papers.
- Refuted/withdrawn: D18.
- Cite-grade audit: `D2_literature_verification.md` (Cranston’s three pairs; Kostochka–Yancey edge bounds 588/638/650; BK crossing threshold 6.77 n vs. Cranston’s 6.95 n).

4. `directed_path_minimum_outdegree/`

Conjecture

Cheng–Keevash Conjecture 1 (attributed to Thomassé). Every oriented graph with minimum out-degree n contains a directed simple path of length 2 (length = number of arcs; vertex-simple).

Status

Active; recent significant closures.

	Status
1–2	Closed (trivial / Cheng–Keevash 2024)
3	Closed , hand proof, all $n \geq 7$
4, $n = 10$	Closed , hand + computer-aided; audited
4, $n = 11$	Closed , computer-aided; two independent pipelines agree
4, $n = 12$	Open; miner ready, computationally demanding
5	Open

For $n = 4$, $n = 10$: Cheng–Keevash Lemma 7 + forced-arc derivation + exhaustive enumeration on the (2,2,1,1) score profile. For $n = 11$: 32 configurations, 117.9 M completions, zero obstructions.

Layout

- `docs/` — proof manuscripts (`k3_hand_proof.md`, `k4_n10_proof.md`, `k4_n11_proof.md`, `k4_partial_appendix.md`), `literature_notes.md`, `plan.md`.

- `data/` — audit outputs, certificates, path tables (e.g., `k4_n10_certificate.json`, 10 MB).
- `scripts/` — miners (`k4_local_miner.py`, `k4_score_profile_miner.py`), independent re-implementations (`k4_score_profile_independent_check.py`), verifiers (`k4_verify_certificate.py`, `verify_directed_path_counterexample.py`).

Start here

1. `README.md` — status table + reproducer commands.
2. `docs/plan.md` (attack plan, status audit, verified literature facts).
3. `docs/k3_hand_proof.md` — simplest closed case; Lemma 7 + cyclic-closure technique.
4. `docs/k4_n10_proof.md` (theorem + reductions).

Notable artifacts

- Lemma 7 + oriented average bound (core structural tool from Cheng–Keevash 2024).
- Two independent miners agree on all 24 (S,T) configurations / 3,664 completions for $n = 10$.
- Mechanical certificate verifier: hash, 4-outregularity, forced-arc compliance, witness paths.
- `k4_score_profile_miner.py` 12 is the next computational target (~2–4 h).

5. earth_moon_problem/

Conjecture

Ringel (1959). Define a graph as *biplanar* if its edges can be partitioned into two planar subgraphs. What is the maximum chromatic number χ_{EM} of a biplanar graph? Known bounds: $9 \leq \chi_{EM} \leq 12$ (Sulanke/Gardner 1980 lower; Heawood 1890 upper). The folklore $\chi_{EM} = 9$ is unproven.

Status

Open. Two complementary tracks:

- **Disprove $\chi_{EM} = 9$** by exhibiting a 10-chromatic biplanar graph. Headline candidate C (28 vertices, $\chi = 10$, K_9 -free) ruled **UNSAT** after 9.7 h (Phase 4, May 2026).
- **Tighten $\chi_{EM} = 11$** via density: K_9 -free 12-critical biplanar graphs need $(65n - 54) / 11$ edges by Kostochka–Yancey, but biplanarity caps them at $6n - 12$. The gap closes for $n \geq 78$; further K_9 -free strengthening would tighten it.

Recent (commit 204282a, May 16): Fork C probe — a non-saturated K_7 witness on partition (7,3,1) (14 vertices, 56 edges) is biplanar in 0.011 s, even

though $K_7 + K_5$ is not. This blocks one Q0-closure path; the discharge argument now branches to Forks A/B (structural Gallai arguments on low-degree vertices).

Layout

- `docs/` — Phase 6 discharge attempt (50+ pages), literature notes, plan, upper-bound audit, paper skeleton.
- `scripts/` — biplanar oracle (SMS wrapper), profile enumerator, cycle-blowup generator, SAT runner.
- `data/` — test runs: `c7_k4` (28 V, 9.7 h UNSAT), `candidate1/2`, `new_c5_a/b`.
- `external/sat-modulo-symmetries` — KSS 2023 SAT-modulo-symmetry framework (submodule).

Start here

1. `README.md` — 2-minute problem statement and status.
2. `docs/plan.md` — Phase structure (oracle, regression, mining, upper bounds).
3. `docs/phase6_discharge_attempt.md` — current frontier: 89-vertex 12-critical structure, Gallai forest on low vertices, the Fork C K_7 finding.
4. `data/c7_k4/README.md` — how C K was ruled out; scaling vs. C cases.

Notable artifacts

- `biplanar_check.py` — flexible biplanarity tester (SMS wrapper).
- `phase6_discharge_attempt.md` — sections 7.15–7.16 document Forks A/B/C.
- Fork C result: $K_7 + 7$ cyclic 5-neighbours ($n = 14$, $m = 56$) is biplanar.
- `c7_k4` run: 34,834 CPU s, 64 M propagator calls, 3.84 M Kuratowski clauses.

6. pebbling_cartesian_product/

Conjecture

Graham’s pebbling conjecture. For connected graphs G, H : $(G \times H) \leq (G) \cdot (H)$. This project targets the **Lemke square** $L_{\text{fpy}} \leq L_{\text{fpy}}(L) = |V(L)| = 8$, the classical stress test that would yield a 64-pebble counterexample if the conjecture fails on it. Rational weight-function certificates via column generation produce upper bounds.

Status

Terminal — clean stopping point. Four verified deliverables:

- Global bound: $(L_{\text{fpy}} \leq L_{\text{fpy}}) \leq 246$ (rational, re-checkable).

- Rooted: $(\cdot, (v, v))$ 106 (sharpens Hurlbert 2017’s 108).
- Negative pricing: the bottleneck orbit (0,0) is LP-optimal under two bounded strategy classes; no improving column found.
- FPY ingestion bridge with current Flocco–Pulaj–Yerger source works; reproducing their published 96 bound is blocked on Gurobi or author JSON (missing dual multipliers; CSV ambiguity vs. published artefacts).

No FPY 96 reproduction or 64-pebble counterexample search — scope limits documented in the terminal report.

Layout

- docs/ — terminal_report.md (entry point), phase2b_status.md (FPY blocker), lp_improvement_log.md, literature_notes.md, plan.md.
- data/pebbling_product/ — 22 root-orbit certificates (JSON), root_orbit_bounds.csv, known_bounds.csv, L_fpy.json.
- scripts/ — 19 Python tools: exact verifier, rational checker, LP harness, column-generation oracles (paths, Y-tree, trident, II-tree), FPY adapter, orbit aggregator.
- tests/ — 16 pytest modules (101 tests, all passing).
- external/ — gitignored FPY clone directory.

Start here

1. docs/terminal_report.md — headline results, comparison table, FPY blocker.
2. README.md — 2-minute orientation; reproduction commands.
3. scripts/check_pebbling_weight_certificate.py — the verifiable core (~700 lines).
4. data/pebbling_product/root_orbit_bounds.csv — 22 orbits, bounds 68–246.

Notable artifacts

- Hurlbert_path_augmented_v1v1_1e106.json — sharpened rooted bound.
- path_orbit_0_0_max_len7.json — bottleneck certificate (orbit (0,0) at 246).
- price_tree_strategy.py — pricing oracle, 196 M+ nodes explored at depth 7.
- fpy_probe.py — live FPY round-trip self-test.

Last updated 9 May 2025. Self-contained, independently verified; future work options: (i) run FPY’s MILP with Gurobi, (ii) widen the priced strategy class, (iii) leave 246 as standalone.

7. positive_square_energy_equality/

Conjecture

Akbari–Kumar–Mohar–Pragada–Zhang Conjecture 9.2 (arXiv:2506.07264, June 2025). For a connected graph G of order n :

- (i) $s(G) = n - 1$ iff G is a tree.
- (ii) $s(G) = n - 1$ iff G is a tree or a complete graph K_n .

Here $s_{\pm}(G)$ is the sum of squares of the positive (resp. negative) adjacency eigenvalues. Refines Elphick–Farber–Goldberg–Wocjan ($s_{\pm} = n - 1$).

Status

Conjecture 9.2 is open in general. For **2-trees** (maximal chordal, treewidth 2) substantial progress is in:

- **Unconditional proofs.** Books B_k : $(B_k) = 2 - 4 / (\sqrt{8k+1} + \sqrt{8k-7})$. 2-paths asymptotic: $\mu(L) = (32 - 27\sqrt{3}) / (12)$ 1.4262 per vertex (Szegő–Cesàro via Stieltjes transforms, Phases 10–11). Lemmas B1 and B1 (Rayleigh bounds).
- **Finite- n rigorous.** $(L_n) = 17/16$ for $n \geq [4, 2000]$ via Demmel–Kahan a-posteriori certificates.
- **Headline open.** “(L’) existential ear-selection lemma” — does every 2-tree admit a simplicial degree-2 vertex with $\pm 17/16$?
- **The wall.** Condition (b) — the slot-shift sum bound (O12.2). Estimated 6 person-months to 2 years.

Layout

- **paper/** — 40-page LaTeX submission draft (7 sections + appendices); `\stub{}` markers for unfilled proofs.
- **docs/** — 30 technical notes: phase logs, failure-mode catalogues (F1–F15), structural analyses of books / 2-paths / bad-ears.
- **scripts/** — corpus builders, moment calculations, Stieltjes transforms, Demmel–Kahan verification.
- **tests/** — 14 regression test files (519 passing tests).
- **data/** — JSON fixtures: 2235-record max-degsum census, 1795-graph corpus, 1063+ graph-family database.

Start here

1. **README.md** — executive summary; verdict after Phase 12 reviews.
2. **docs/plan_v14.md** — strategic map of what works, what failed, why.
3. **paper/paper.pdf** (rebuild via `make` in **paper/**).
4. **docs/corollaries_AB.md** — two short corollaries (claw-free, diameter 2).

Notable artifacts

- Phase 8 Lemma B1 + Phase 12.A Lemma B1 (Rayleigh bounds).
- Phase 10–11 asymptotic: $\lim_n I(L_n, v^*) = I_\infty(L)$ 1.0157 via spectral measure theory.
- Books closed form; 2-paths asymptotic via half-line Stieltjes + Portmanteau.
- Failure modes (Appendix B): F14 (trace-identity reformulation does not decouple the slot-shift wall); F15 (Theorem 8.1 hypothesis empirically inapplicable to 2-trees).
- Workstream has pivoted from research-attack to paper-writing.

8. unit_vector_flows/

Conjecture

A **unit vector flow** assigns to each edge of a graph a unit vector in $\mathbb{S}^3$ (a point on $\mathbb{S}^2$) such that the oriented sum of incident vectors at every vertex is zero (Kirchhoff conservation). **Jain’s surviving conjecture** (2007):

Every bridgeless graph admits an $\mathbb{S}^2$ -flow.

A related quaternion-flow conjecture was recently disproved by Ulyanov (2026).

Status

A finite theorem, rigorously proved by computer assistance:

Theorem. Every nontrivial snark on at most 28 vertices admits an $\mathbb{S}^2$ -flow.

All 3,247 nontrivial snarks $n \leq 28$ carry independently replayable interval-Krawczyk certificates: numerical witness (Levenberg–Marquardt on the polynomial Kirchhoff system) + 50-digit interval verification.

Jain’s conjecture itself remains open for general bridgeless graphs.

Preprint finished (`paper/main.tex`, 5 sections); covers algorithm, enumeration, and negative results on symmetric flower-snark constructions.

Layout

- `docs/` — `plan.md` (phases 1–4), literature review, flower-snark analysis, cycle-double-cover obstruction notes.
- `scripts/` — `catalogue.py` (snark enumeration + SAT filter), `sweep.py` (numerical witness), `interval.py` (Krawczyk certifier), `verify_sweep.py` (replay verifier).
- `data/` — 3,247 snark catalogue (g6), witness JSON, 3,247 interval certificates (schema v2), manifest with SHA-256 provenance.

- `tests/` — 41 regression tests; negative calibration confirms rejection of bridged graphs.
- `paper/` — LaTeX preprint (intro, preliminaries, certification pipeline, finite theorem, CDC obstructions).

Start here

1. `THEOREM.md` — formal statement, enumeration provenance, Krawczyk algorithm, replay protocol.
2. `README.md` — context, headline table, reproduction instructions.
3. `docs/plan.md` — phase roadmap.
4. `paper/main.tex` — full certification algorithm and limitations.

Notable artifacts

- Theorem statement: `THEOREM.md` lines 5–26.
- Makefile targets: `verify`, `verify-certs`, `catalogue`, `sweep`, `certs`.
- Certificate manifest: `data/catalogues/nontrivial_snarks_n10_to_28.manifest.json`.
- Replay verifier: `scripts/verify_sweep.py` (recomputes Krawczyk from scratch).
- Negative result: `docs/flower_snarks.md` (symmetric constructions fail for $J_{\{2k+1\}}$).
- Quick reproduction: `make -C problems/unit_vector_flows verify-certs` (~70 min).